Introduction to BERT

Setting up the environment

```
!pip install transformers torch --quiet
```

```
import pandas as pd
from transformers import pipeline # used to import the Hugging Face transformers library pre-trained models
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
```

A pre-trained sentiment pipeline

```
# Load the BERT-based sentiment analysis pipeline
sentiment_model = pipeline(task="sentiment-analysis", model="nlptown/bert-base-multilingual-uncased-sentiment")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%                                                   953/953 [00:00<00:00, 54.8kB/s]
model.safetensors: 100%                                             669M/669M [00:06<00:00, 151MB/s]
tokenizer_config.json: 100%                                         39.0/39.0 [00:00<00:00, 1.56kB/s]
vocab.txt:        872k/? [00:00<00:00, 18.5MB/s]
special_tokens_map.json: 100%                                       112/112 [00:00<00:00, 7.52kB/s]
Device set to use cpu
```

```
# quick test
sentiment_model("This phone case is simply incredible")
```

```
[{'label': '5 stars', 'score': 0.9230801463127136}]
```

BERT sentiment analysis on example text

```
reviews = [
    "Shipping was super fast and customer support was helpful.",
    "The product stopped working after only two weeks. Very disappointed.",
    "It's okay, not great but not terrible either."
]
```

```
results = sentiment_model(reviews)
for text, result in zip(reviews, results):
  print(f"TEXT: {text}\nPREDICTION: {result}\n")
```

```
TEXT: Shipping was super fast and customer support was helpful.
PREDICTION: {'label': '5 stars', 'score': 0.6436988711357117}

TEXT: The product stopped working after only two weeks. Very disappointed.
PREDICTION: {'label': '1 star', 'score': 0.7773727774620056}

TEXT: It's okay, not great but not terrible either.
PREDICTION: {'label': '3 stars', 'score': 0.8755016922950745}
```

BERT sentiment analysis of online reviews

```
import pandas as pd
df = pd.read_csv("videogame_reviews.csv") # amzn_reviews.csv has a column 'review_text'
df.head()
```

	product_id	reviewer_id	review_title	star_rating	review_date	review_text
0	B0009SQFHA	AFFRMIDDNSX4DVEQBKBX2VKEGCYA	Condemned is a scary fun game, This is the gam...	5	6-Apr-16	Condemned is a scary fun game, This is the gam...
1	B0009SQFHA	AEHFBP6OLECUQA3YP625PZNYQIOQ	One of the original launch release titles!	5	19-Jun-16	Supposed to be one of the scariest games on th...
2	B0009SQFHA	AHSS4VDFPOZVDXEHAVEMCBVZVQUA	scary game with more than just jump scares.	5	27-Jun-16	I loved playing this. Being a fan of horror ga...
3	B0009SQFHA	AEFWMXAPQBJO7BYO35VB3BHCW75Q	Not fun. Lacking substance.	2	29-Jun-16	Even though I'm a sucker for cop games, this i...
4	B0009SQFHA	AFWHPV62GV2QVRRTY3WTUNAIVBIQ	Limited edition game, recommend buying it whil...	5	15-Jun-17	The disk was in perfect condition! There's no ...

```
# For a quick demo, let’s sample a subset
sample_df = df.sample(50, random_state=12)

sample_df["bert_sentiment"] = sample_df["review_text"].apply(
    lambda x: sentiment_model(str(x))[0]["label"]
)

sample_df.head()
```

	product_id	reviewer_id	review_title	star_rating	review_date	review_text	bert_sentiment
9593	B005FYJA52	AFRUYVZCQ2HFHX5BDOFNXYERRUYQ	Just when you dont think it can get any more a...	5	5-Feb-16	Love Super Mario Galaxy. I have Super Mario G...	4 stars
2883	B002BRZ9G0	AE2HOUIKG3AYAIBZJFJILJ3DA2HQ	Five Stars	5	20-May-18	It work great in my will. My kids love it. Tha...	5 stars
2629	B002BRZ79E	AGA6H5Z2UZU76BAUVKJIMNSGOCDQ	Great game, Great Final Fantasy game	5	25-Dec-16	This is a great follow up to the Final Fantasy...	5 stars
11562	B0086V5V9A	AGTXM4EVQQBSAQS2INOEYN6IM6YA	Fantastic game, loved it	5	11-Apr-16	Fantastic game, loved it. Lot's of innovation...	5 stars
196	B000FRU0NU	AEXWKVSLGBMZRKWGSKMZC4FFN77Q	Sucky game	1	13-Aug-16	Doesn't work makes my xbox freeze and restart....	1 star

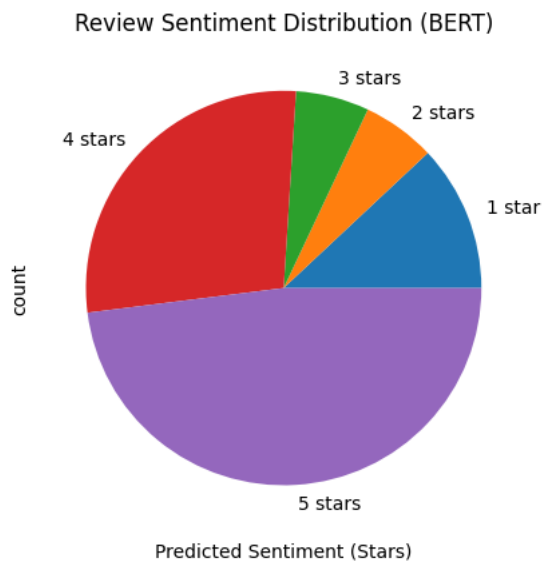
Aggregating sentiment results

```
sentiment_counts = sample_df["bert_sentiment"].value_counts().sort_index()
print(sentiment_counts)

# Convert counts to percentages
sentiment_pct = sentiment_counts / sentiment_counts.sum() * 100
print(sentiment_pct)
```

```
bert_sentiment
1 star      6
2 stars     3
3 stars     3
4 stars    14
5 stars    24
Name: count, dtype: int64
bert_sentiment
1 star     12.0
2 stars     6.0
3 stars     6.0
4 stars    28.0
5 stars    48.0
Name: count, dtype: float64
```

Visualizing sentiment distribution

```
import matplotlib.pyplot as plt

sentiment_counts.plot(kind="pie")
plt.xlabel("Predicted Sentiment (Stars)")
plt.title("Review Sentiment Distribution (BERT)")
plt.show()
```



BERT for embedding and clustering

BERT as an embedding model

```
# Import needed libraries
from transformers import AutoTokenizer, AutoModel

# Set up the tokenizer and model
model_name = "bert-base-uncased"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModel.from_pretrained(model_name)

# Define a function to do CLS (classification) embedding using the tokenizer and model
def get_cls_embedding(text):
    inputs = tokenizer(
        text,
        return_tensors="pt",
        truncation=True,
        padding=True,
        max_length=128
    )
    with torch.no_grad():
        outputs = model(**inputs)
    # CLS token embedding is at position 0
    cls_embedding = outputs.last_hidden_state[:, 0, :].squeeze().numpy()
    return cls_embedding
```

```
tokenizer_config.json: 100%                 48.0/48.0 [00:00<00:00, 1.98kB/s]
config.json: 100%                           570/570 [00:00<00:00, 19.3kB/s]
vocab.txt: 100%                             232k/232k [00:00<00:00, 5.86MB/s]
tokenizer.json: 100%                        466k/466k [00:00<00:00, 10.1MB/s]
model.safetensors: 100%                     440M/440M [00:11<00:00, 74.5MB/s]
```

Clustering reviews based on BERT embeddings

```
from sklearn.cluster import KMeans
import torch
import numpy as np

# Take the small subset of reviews we defined earlier
texts = sample_df['review_text'].tolist()

# Do embedding for all the reviews and stack them
embeddings = np.vstack([get_cls_embedding(t) for t in texts])

# Do Kmeans clustering based on the embeddings
kmeans = KMeans(n_clusters=3, random_state=12)
labels = kmeans.fit_predict(embeddings)

for text, label in zip(texts, labels):
    print(f"Cluster {label}: {text[:80]}...") # print the first 80 characters of texts and their cluster
```

```
Cluster 2: Love Super Mario Galaxy.  I have Super Mario Galaxy 2 now and it is just as grea...
Cluster 0: It work great in my will. My kids love it. Thank you very much. I ma very satisf...
Cluster 1: This is a great follow up to the Final Fantasy XI, an MMORPG.  There were proble...
Cluster 2: Fantastic game, loved it.  Lot's of innovation and well made all around.  The ga...
Cluster 2: Doesn't work makes my xbox freeze and restart. Brandy new never worked right got...
Cluster 2: We were so excited for this but my kid ended up in tears and I ended up incredib...
Cluster 0: I just got the PS4 Ratchet & Clank and loved it.  Cute characters, witty dialogu...
Cluster 0: Probably the best PS Vita game out there and the high reviews on Amazon and ever...
Cluster 2: Awesome game came in perfect condition with no damages to the box or the game in...
```

```
Cluster 0: What more can be said about this game/series.  Every year it improves and this i...
Cluster 2: Bought this for my 6 year old. She loved it, and the game was perfect for her si...
Cluster 0: It is good for kids. Non violent and yet entertaining games for kids...
Cluster 0: I enjoyed the demo and fianlly pulled the trigger on this.  Not disappointed at ...
Cluster 1: This is Fable II, the sequel to Fable and Fable The Lost Chapters/Fable Annivers...
Cluster 2: No scratches. Works great and my so love this game Packing was great as well...
Cluster 2: A phenomenal game. I heard it was great, but never actually bothered to play it....
Cluster 0: This game exceeded all my expectations. The battles get more complex as you prog...
Cluster 0: Thank you very much, my order arrived on time and I'm really happy...
Cluster 2: Love the game. Really nostalgic for me since growing up I had a Nintendo 64. Now...
Cluster 0: Despite the few bugs on the first two games which won't take away the joy of pla...
Cluster 2: Kids seem to really like it!  They say it's a fun game to play!...
Cluster 0: I'm a fairly new PS Vita owner and this is the best game I've played so far.  I ...
Cluster 2: The game is kind of fun! I was expecting it to be way better especially given th...
Cluster 2: Our kids just never got into Wii.  The game is good quality...just didn't fit ou...
Cluster 0: My daughter loves the game, but is somewhat disappointed by the fact that the ch...
Cluster 0: I've loved Mario games since I was a kid. I will forever continue purchasing the...
Cluster 2: I bought Ghost Warrior 2 and I was blown away at how much better it was in terms...
Cluster 2: Don't much care for it on first look...too much dialog in the beginning and the ...
Cluster 1: for a Vita game, for a Vita LEGO game, this is a bit challenging. which is good,...
Cluster 2: Love this game ! Fun to play with kids -nice break from the action games 😊👍...
Cluster 2: Dope game...Main quests do feel like a repetitive chore at times.End game and NG...
Cluster 2: My 5yr old son really enjoys this game, seems pretty age appropriate for him, pl...
Cluster 2: Good game. Lots of fun. Grandson played it for hours when we first got it....
Cluster 2: Even today, five years after it's release, this is just the best Kinect game i h...
Cluster 2: Disc arrived completely scratched. Would not work in Wii player. Very disappoint...
Cluster 0: This game is a pure CLASSIC. The art book and everything else is great for any f...
Cluster 2: Terrible story Ugly Graphics awkward controllsWORSE OF ALL WHEN YOU FAIL A MISSI...
Cluster 0: An RPG experience that will stay with you long after you're done. The fun gamepl...
Cluster 2: language is "rated R" and not always necessary... so 4 stars...but the disc work...
Cluster 1: This game captures the challenges of the in characters in the movie! Although, y...
Cluster 1: This game is fun, yet this game is out of date.If you have Rock Band 4, no, you ...
Cluster 2: This is the game I've played pretty much daily for the last 2-3 years. I bought ...
Cluster 0: Not crazy about this one.  I remember playing DOA for the Dreamcast many moons a...
Cluster 0: I have been playing the game for 2 hours now and I can easily say that I love ev...
Cluster 2: Seller was very honest. Said like new and meant it. game worked from start to fi...
Cluster 2: Granddaughter was thrilled  to received this with her 3DS for Christmas.  Only p...
Cluster 2: Flawless! I'll skip passed the fanboying, and just say this game is a solid 10/1...
Cluster 0: It's awful, but you know that, buy it for a great time, HILARIOUS TIME...
Cluster 0: Good game for fans who still like and miss the NCAA Football franchise....
Cluster 2: Sealed brand new was exactly what I thought it would be perfect condition was el...
```

Continuous sentiment analysis (between -1 and 1) using BERT

```
from transformers import pipeline # we need the pipeline like before

# BERT-family sentiment model (binary: POSITIVE / NEGATIVE)
sentiment_model = pipeline(
    task="sentiment-analysis",
    model="distilbert-base-uncased-finetuned-sst-2-english"
)

# The model above returns postivie, negative labels along with a confidence. We turn it into a score using this function:
def bert_signed_score(text):
    out = sentiment_model(text, truncation=True)[0]
    label = out["label"].upper()      # 'POSITIVE' or 'NEGATIVE'
    score = out["score"]              # confidence in predicted label (0 to 1)
    if label.startswith("NEG"):
        return -score                 # map NEGATIVE to [-1, 0)
    else:
        return score                  # map POSITIVE to (0, 1]

text = "Shipping was super fast and the support team was helpful."
print(text)
print("Sentiment score:", bert_signed_score(text))  # in [-1, +1]
```

```
config.json: 100%                                   629/629 [00:00<00:00, 31.8kB/s]
model.safetensors: 100%                             268M/268M [00:06<00:00, 32.4MB/s]
tokenizer_config.json: 100%                         48.0/48.0 [00:00<00:00, 4.42kB/s]
vocab.txt: 100%                                     232k/232k [00:00<00:00, 5.42MB/s]
Device set to use cpu
Shipping was super fast and the support team was helpful.
Sentiment score: 0.9992145299911499
```

Applying the continuous sentiment scores to reviews

```
# Apply BERT-based continuous sentiment to a DataFrame of reviews
def sentiment_score(text):
    return bert_signed_score(str(text))

sample_df["sentiment_score"] = sample_df["review_text"].apply(sentiment_score)

sample_df.head()
```

	product_id	reviewer_id	review_title	star_rating	review_date	review_text	bert_sentiment	sentiment_score
9593	B005FYJA52	AFRUYVZCQ2HFHX5BDOFNXYERRUYQ	Just when you dont think it can get any more a...	5	5-Feb-16	Love Super Mario Galaxy. I have Super Mario G...	4 stars	0.998959
2883	B002BRZ9G0	AE2HOUIKG3AYAIBZJFJILJ3DA2HQ	Five Stars	5	20-May-18	It work great in my will. My kids love it. Tha...	5 stars	0.999876
2629	B002BRZ79E	AGA6H5Z2UZU76BAUVKJIMNSGOCDQ	Great game, Great Final Fantasy game	5	25-Dec-16	This is a great follow up to the Final Fantasy...	5 stars	0.999371
11562	B0086V5V9A	AGTXM4EVQQBSAQS2INOEYN6IM6YA	Fantastic game, loved it	5	11-Apr-16	Fantastic game, loved it. Lot's of innovation...	5 stars	0.999886

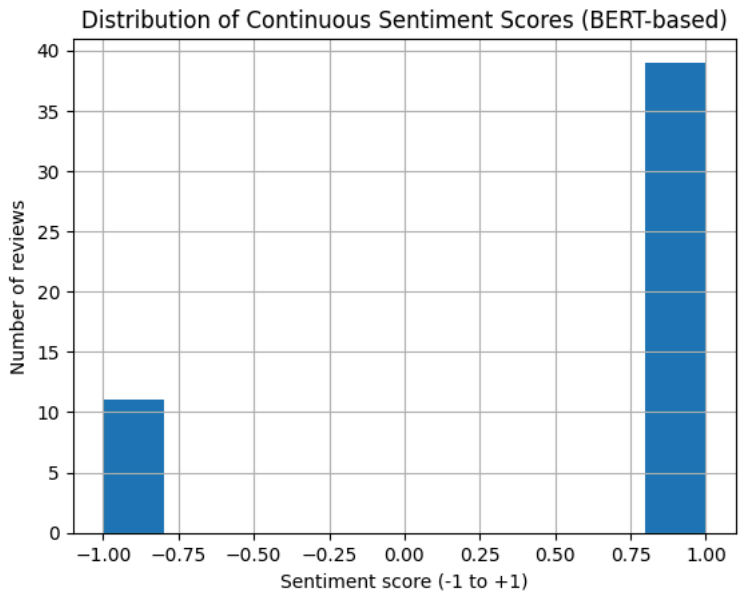
Aggregating continuous scores

```
# Aggregate by time, product, or campaign (example: by 'product_id')
product_sentiment = (
    sample_df
    .groupby("product_id")["sentiment_score"]
    .mean()
    .sort_values()
    )
print(product_sentiment.head(10)) # since we randomly drew 50 reviews, there are not that many reviews per product_ID
```

```
product_id
B004HYK956   -0.999816
B0050SYYTK   -0.999786
```

```
B000FRU1UM   -0.999765
B007H45BSI   -0.999478
B001GIUWX0   -0.999471
B000FRU0NU   -0.999421
B0050SYK44   -0.999310
B000XJNTNS   -0.998614
B001BX6JUA   -0.992706
B002I092MM   -0.988573
Name: sentiment_score, dtype: float64
```

```
# Or simply look at the distribution overall
import matplotlib.pyplot as plt

sample_df["sentiment_score"].hist(bins=10)
plt.xlabel("Sentiment score (-1 to +1)")
plt.ylabel("Number of reviews")
plt.title("Distribution of Continuous Sentiment Scores (BERT-based)")
plt.show()
```




```
# amzn_reviews.csv has a column 'review_text'
df = pd.read_csv("videogames_reviews.csv")

# For a quick demo, let's sample a subset
sample_df = df.sample(100, random_state=12)

sample_df["bert_sentiment"] = sample_df["review_text"].apply(
    lambda x: nlp_sentiment(str(x))[0]["label"]
)

sample_df.head(15)
```

	review_title	star_rating	review_date	review_text	bert_sentiment
11852	Five Stars	5	31-Jan-16	Great	5 stars
162872	Five Stars	5	3-Mar-18	arrived on time, no problems	5 stars
56029	Awesome	5	12-Jul-16	It's so good	5 stars
856	Still Not Getting It	2	3-Jan-16	Every 2 two minutes, there is a cut scene, and...	2 stars
144932	Perfect way to revitalize a classic	5	6-Nov-17	Nintendo did an amazing job updating this old ...	5 stars
148686	AWESOME GAME!!!!	5	5-Dec-17	This "Resident Evil 4 - PlayStation 4 Standard...	5 stars
154730	Five Stars	5	12-Jan-18	Gift	5 stars
111788	Perfect for my collection	4	23-Apr-17	Great item but where is the code for the expan...	3 stars
14353	I should have listened....	2	6-Feb-16	I should have listened to the reviews stating ...	2 stars
46247	Great stealth based game	4	31-May-16	I didn't know this was made by bethesda until ...	4 stars
92821	Five Stars	5	11-Feb-17	Very happy with my item and delivery	5 stars
54829	Five Stars	5	6-Jul-16	My wife enjoys this game as part of her workou...	4 stars
139397	Five Stars	5	28-Sep-17	Grandson loved this game for his play station....	5 stars
152141	Five Stars	5	28-Dec-17	EXCELLENT GAME	5 stars
131577	Overall the combat is fun, fluid	4	9-Aug-17	If you are a fan of the 'Dynasty Warriors' sty...	4 stars

```
# Apply BERT-based continuous sentiment to a DataFrame of reviews
def sentiment_score(text):
    return bert_signed_score(str(text))

sample_df2 = df.copy()

sample_df2["sentiment_score"] = sample_df2["review_text"].apply(sentiment_score)

sample_df2.head(20)
```

	product_id	reviewer_id	review_title	star_rating	review_date	review_text	sentiment_score
0	B0009SQFHA	AFFRMIDDNSX4DVEQBKBX2VKEGCYA	Condemned is a scary fun game, This is the gam...	5	6-Apr-16	Condemned is a scary fun game, This is the gam...	0.999369
1	B0009SQFHA	AEHFBP6OLECUQA3YP625PZNYQIOQ	One of the original launch release titles!	5	19-Jun-16	Supposed to be one of the scariest games on th...	0.999672
2	B0009SQFHA	AHSS4VDFPOZVDXEHAVEMCBVZVQUA	scary game with more than just jump scares.	5	27-Jun-16	I loved playing this. Being a fan of horror ga...	0.998721
3	B0009SQFHA	AEFWMXAPQBJO7BYO35VB3BHCW75Q	Not fun. Lacking substance.	2	29-Jun-16	Even though I'm a sucker for cop games, this i...	-0.999585
4	B0009SQFHA	AFWHPV62GV2QVRRTY3WTUNAIVBIQ	Limited edition game, recommend buying it whil...	5	15-Jun-17	The disk was in perfect condition! There's no ...	0.988626
5	B0009SQFHA	AGJ2GGXLL4HACHXVBEF5MCRAEXFA	Scam alert	1	21-Jan-18	I can't get passed the first 5min. No dead bod...	-0.999614
6	B0009SQFHA	AHWSGPWMZZU5SVQDDRDNMAJ5TYQA	Best 10 bucks ever 10/10	5	9-Oct-18	Honestly I bought it for like 10 bucks thinkin...	-0.998637
7	B0009Z3HOW	AHXYH4HOMLCCCBGI5VQ6KNHMBDGQ	Great game t This is a Tony Hawk game better ...	5	26-Nov-16	Great game tThis is a Tony Hawk game better th...	0.998792
8	B0009Z3HOW	AFK3DQPDAUZZPFHELDBO6WPFJLXQ	It’s great	5	11-Aug-17	I got a used copy and for used it almost looks...	-0.990257
9	B0009Z3HOW	AGQK34ULQ5OU5OZJAGSWBTAUP3AQ	it doesnt work!!	1	11-May-18	defiantly not a good buy for me the game came ...	-0.998863
10	B0009Z3HOW	AE356RZLTZB4E7265PDHRGNJ3G3A	It’s for certain people	5	27-Jun-18	Now for a tony hawk game I wasn’t expecting mu...	-0.996532
11	B000B6ML0A	AFLL7EN45WOCFOACYETLXZX66WAA	Excellent quality.	5	1-Sep-17	This game was a personal, guilty pleasure of m...	0.999658
12	B000B6ML0A	AFB7RBBWUOUCVCICIYECDTEHS5AQ	Five Stars	5	10-Dec-17	I loved playing this game when I was younger! ...	0.999860
13	B000B6ML0A	AH5ERK7ELAZYVPT5QZXKGQYU2SXA	The item was not platinum hits like the descri...	3	17-Apr-18	The item was not platinum hits like the descri...	-0.914918
14	B000B6ML1Y	AHXVKWTXPNZYTK7QL3JV7PEXHWHA	Simple fun is the best kind of fun	5	18-Feb-16	This is a great game at a great price. The ab...	0.999862
15	B000B6ML1Y	AHV45KKLONNDE3HCFZAZJDFJEXQQ	Pointless. Tons of glitches. Worthless buy Gam...	1	8-Mar-16	Load of crap worst game I've ever played I wou...	-0.999754
16	B000B6ML1Y	AGEOZUNA6MFTFNCLDWPGWLAUZJEQ	and I'm upset it didn't register with regular ...	5	2-Apr-18	The game is awful, and I'm upset it didn't reg...	0.996941
17	B000B6ML28	AHSXRBTWEOBNZ6XZL4BYAF4PZONA	old but very fun game. no one is online	5	12-Aug-16	old but very fun game.no one is online. So unl...	0.998078
18	B000BI3QK2	AHJNV5LX4T6LBCW7ASGT5276EDFA	Five Stars	5	24-Feb-16	Amazing price for an amazing game. Works like ...	0.999869
19	B000BI3QK2	AH5YSVQHFIC7W5TVLIQNKPBOIXIA	Great game, awesome story and quests	5	21-Sep-16	Great game,awesome story and quests,I would re...	0.999886